Vector in C++ → array which grows.

↓
dynamic array

Regular Array → fixed size

(Lecture 20)

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Earners

DSA in C++

Summary Notes

Vector in C++ (how it works internally)

s	c	code
		`vector<int> v;`
0	0	`v.push_back(5);`
1	1	`v.push_back(9);`
2	2	`v.push_back(0);`
3	4	`v.push_back(2);`
4	4	`v.push_back(8);`
5	8	`v.push_back(4);`
6	8	`v.push_back(6);`
7	8	`v.push_back(1);`
8	8	

5

5	9

5	9	0	2

v

5	9	0	2	8	4	6	1

size, capacity

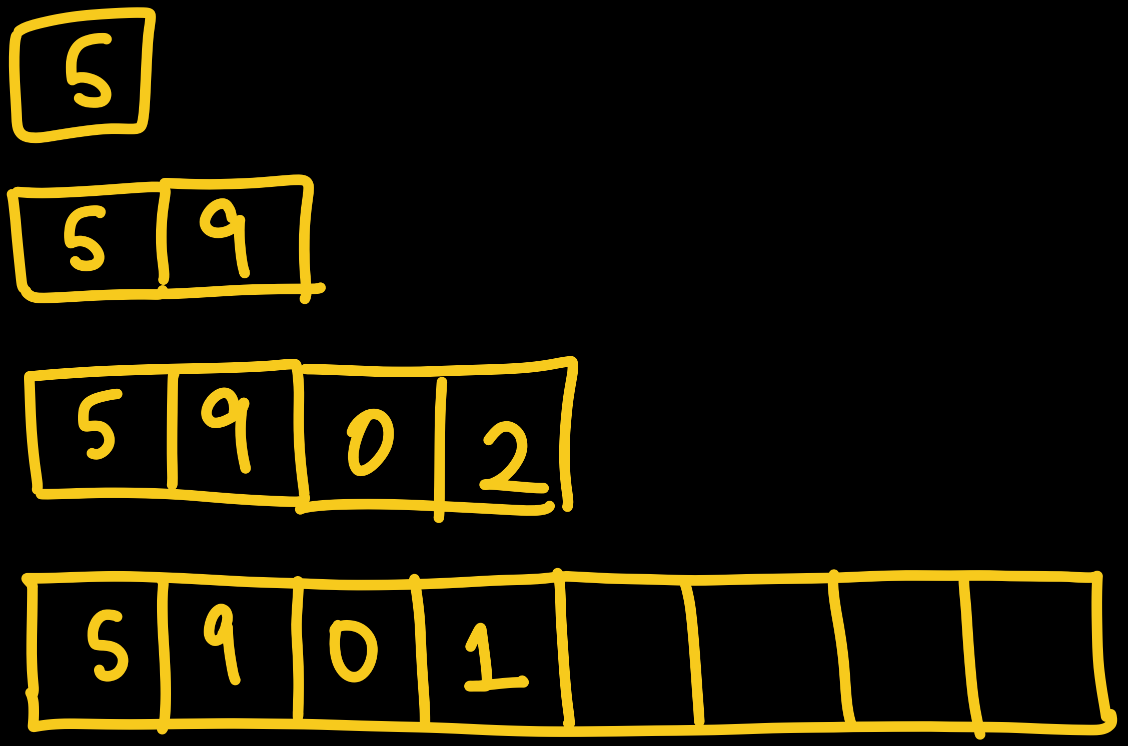
$\boxed{TC}$

$$1+2+3+4+\cdots n \approx \frac{n^2}{2}$$

$$1+2+4+8\cdots n \approx 2n$$

Vector in C++

s	c	
		`vector<int> v;`
0	0	
1	1	`v.push_back(5);`
2	2	`v.push_back(9);`
3	4	`v.push_back(0);`
4	4	`v.push_back(2);`
5	8	`v.push_back(8);`
4	8	`v.pop_back();`
3	8	`v.pop_back();`
4	8	`v.push_back(1);`



Basic operations on Vectors

- syntax
- push_back, pop_back, size, capacity, at, sort

For Each Loop

Passing vectors to functions

Ques: Reverse Array

i

arr = { 70, 60, 50, 40, 30, 20, 10 }

j

5:15
tak
~~break~~
resume

i = 0
j = n-1

```
while(i<j){
    int temp = arr[i]
    arr[i] = arr[j]
    arr[j] = temp
    i++
    j--
}
```

TC = O(n)

Ques: Rotate Array

```
0                 n-1
nums = [1,2,3,4,5,6,7], k = 3
```

↓ rev

0 k-1 k n-1

$$\underbrace{7\ \ 6\ \ 5}_{k}\ \ \underbrace{4\ \ 3\ \ 2\ \ 1}_{n-k}$$

rev ↓ ↓ rev

$$\underbrace{5\ \ 6\ \ 7}\ \ \underbrace{1\ \ 2\ \ 3\ \ 4}$$

Time = $O(n)$

Leetcode 189

Ques: Rotate Array

arr = { 8, 0, 5, 1 }

$K = 70$

$\downarrow$

$K = K \% n$

$70 = 4q + r$

$4 \overline{)70(} q$

$\overline{r}$

Leetcode 189

PW Earne₹s

Ques: Segregate 0s and 1s

M-1

arr[] = [0, 1, 0, 1, 0, 0, 1, 1, 1, 0]

zeros = $\cancel{0}$ $\cancel{1}$ $\cancel{2}$ $\cancel{3}$ $\cancel{4}$ 5

ones = $\cancel{0}$ $\cancel{1}$ $\cancel{2}$ $\cancel{3}$ $\cancel{4}$ 5

Ques: Segregate 0s and 1s Method-2

[0, 1, 0, 1, 0, 0, 1, 1, 1, 0]

0 0 0 0 0 1 1 1 1 1

i

j

Ques: Two Sum

Try Karo

Leetcode 1

Ques: Missing in Array

Try Karo

Leetcode 268

Ques: Wave Array

Try karo